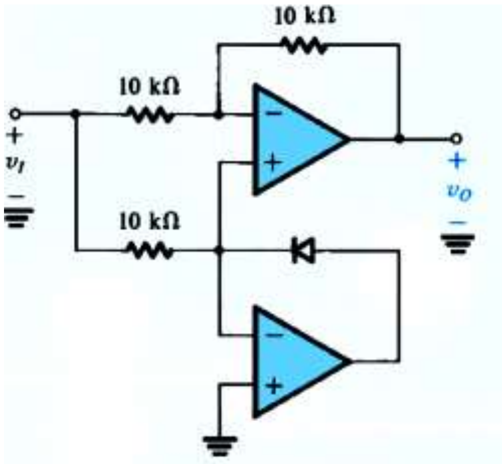
National University of Computer & Emerging Sciences, Lahore
Department of Electrical Engineering (Fall 2024)

EE2001 –Electronic Circuit Design
Assignment # 3 (Total = 30)
CLO-04

Due Date: 28th Nov, 2024

Question 1

Sketch the transfer characteristics of the following circuit.



Question 2

a. For an astable multi-vibrator using 555 timer IC with a C = 1.0 nF capacitor, find the values of R_A and R_B that results in an oscillation frequency of 100 kHz and a duty cycle of 75%. Draw the circuit diagram as well.

b. What is the duty cycle and oscillation frequency of the circuit designed in part (a), that is R_A, R_B and C are the same as in (a); if the a diode is connected in parallel with R_B to bypass it during the charging phase.

Question 3

A pressure sensor outputs a voltage signal P_s ranging from to -2.4 to 1.1 V. For interface to an analog-to-digital converter, this signal is normalized to signal Pm that ranges between 0 to 2.5 V. Find the transfer function equation of P_s to P_m, and hence develop the required signal conversion circuit with high-input impedance.

ECD

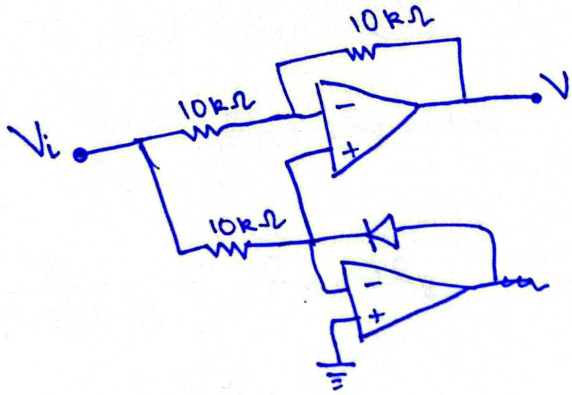
A03

Owais Rao

22L-7638

BEE-5A

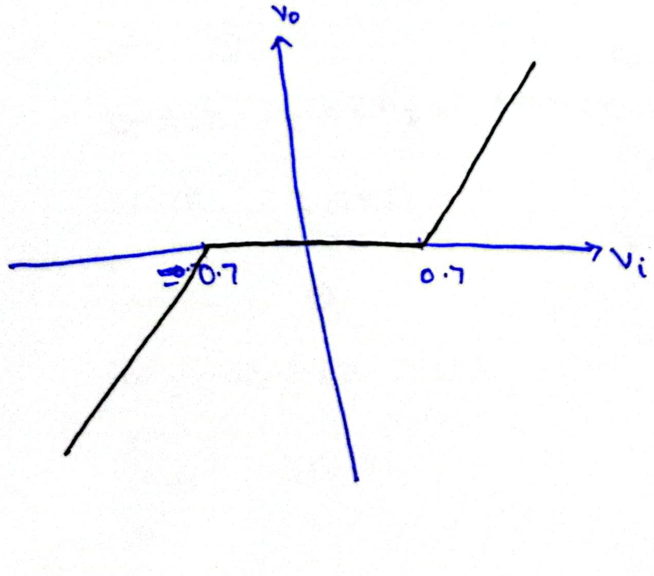
Q.1



When $V_i < -0.7V$,
Diode is reverse biased

When $-0.7V \leq V_i \leq 0.7V$
The input voltage is less than the diode voltage (0.7V), which restricts the output voltage.

When $V_i > 0.7V$,
Diode is forward biased



Q·No·2

(a)

A-Stable MVB using 555 timer

$C = 1nF$, $f = 100\,kHz$, $D = 75\%$

$\Rightarrow T = \frac{1}{f} = \frac{1}{100k} = 10\mu s$

$\Rightarrow T = \ln 2\,(R_A + 2R_B)\,C$

$\frac{10\mu}{\ln 2} = (R_A + 2R_B)(1n)$

$\frac{10\mu}{(\ln 2)(1n)} = R_A + 2R_B$

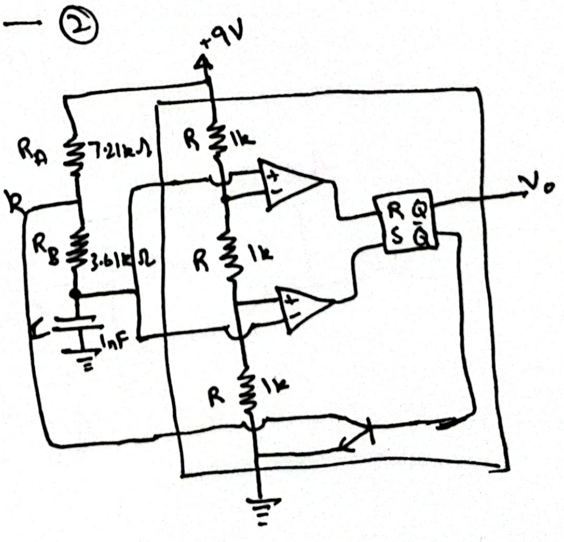
$R_A + 2R_B = 14.43k$ — ①

$\Rightarrow D = \frac{R_A + R_B}{R_A + 2R_B}$

$75\% = \frac{R_A + R_B}{14.43k}$

$10.82k = R_A + R_B$

$R_A = 10.82k - R_B$ — ②



Put in ①

$10.82k - R_B + 2R_B = 14.43k$

$\boxed{R_B = 3.61\,k\Omega}$

Put in ②

$R_A = 10.82k - 3.61k$

$\boxed{R_A = 7.21\,k\Omega}$

(b)

When a diode is connected parallel to R_B, the charging cycle depends on R_A only while discharging cycle depends on R_B.

$$\Rightarrow \quad D = \frac{T_H}{T} = \frac{T_H}{T_H + T_L} = \frac{R_A}{R_A + R_B}$$

$$= \frac{7.21k}{7.21k + 3.61k}$$

$$\boxed{D = 66.6\ \%}$$

$$\Rightarrow \quad T = (\ln 2)(R_A + R_B)\,C$$

$$\frac{1}{f} = (\ln 2)\,(7.21k + 3.61k)\,(1n)$$

$$\frac{1}{7.499 \times 10^{-6}} = f$$

$$\boxed{f = 133.35\ kHz}$$

Q. No. 3

P_s : $-2.4V$ to $1.1V$ $\qquad$ P_m : $0V$ to $2.5V$

As $\quad y = mx + c$

Put $\quad y = P_m$ (output)

and $\quad x = P_s$ (input)

Slope : $m = \dfrac{\text{Output range}}{\text{Input range}}$

$$= \frac{2.5 - 0}{1.1 - (-2.4)} = \frac{2.5}{3.5}$$

$$= 0.714$$

$$\Rightarrow \quad P_m = 0.714 P_s + c$$

For c (offset), Put boundary points i.e $P_m = 0$ & $P_s = -2.4$

$$0 = 0.714(-2.4) + c$$

$$c = 1.7136$$

Hence, the transfer function is

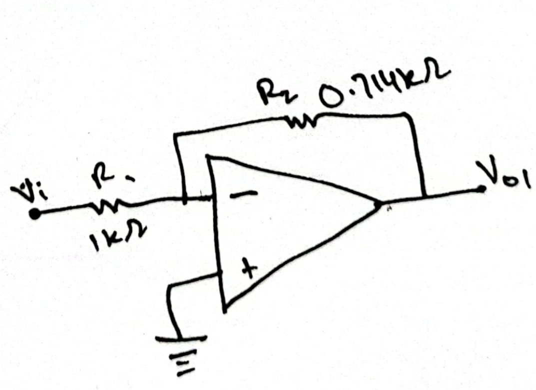
$$\boxed{P_m = 0.714 P_s + 1.7136}$$

For $G = 0.714$,

Using inverting configuration,

$$G = -\frac{R_2}{R_1}$$

$$0.714 = -\frac{R_2}{R_1}$$



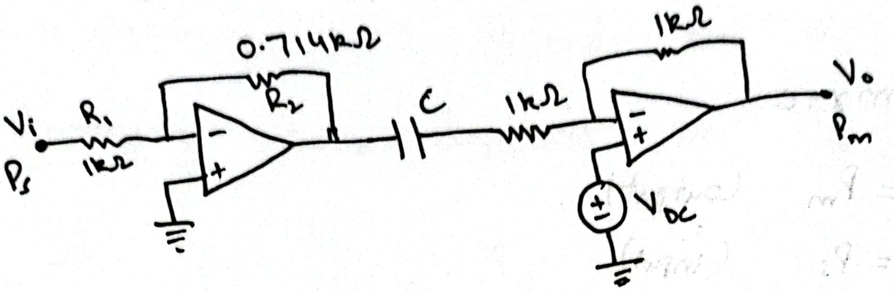
Let $\boxed{R_1 = 1k\Omega}$,

$$(0.714)(1k) = R_2$$

$$R_2 = 714\,\Omega$$

$$\boxed{R_2 = 0.714\,k\Omega}$$

To add offset 1.7136, using summing amplifier configuration with a capacitor.



Here $V_i = P_s$, $V_o = P_m$, $V_{DC} = 1.7136V$

$$\boxed{P_m = 0.714 P_s + 1.7136}$$